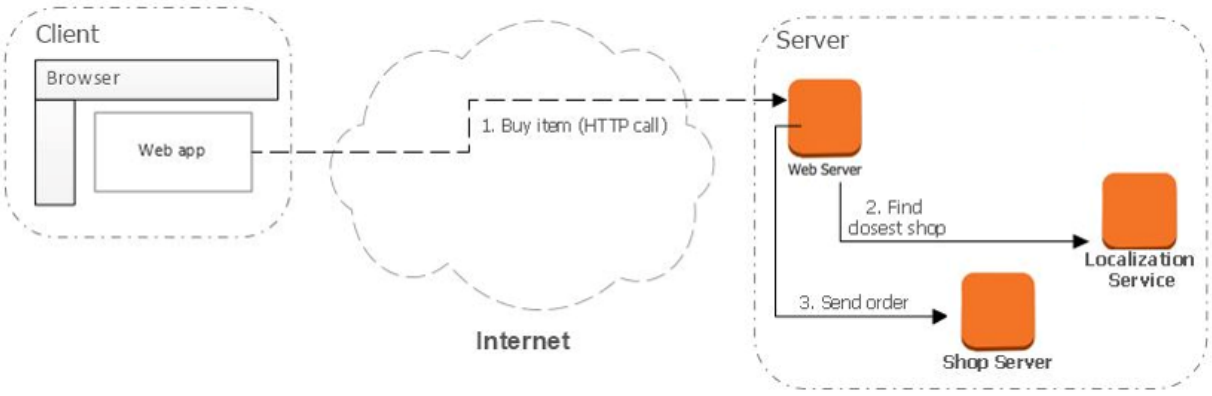


Figure 11.3 Simplified architecture for the e-commerce application described in chapter 8 and here in section 11.1.

The first issue we have overlooked so far is that not all the retailers have all the goods in stock. So, it's not enough to find the closest shop to a customer, we need to find the closest retailer that can actually sell and deliver the product bought by our user. If the user buys more than one item and we need to deliver them all in one go (to save on shipment or just to reduce user's churn) we likely want to filter shops that have all the items at the same time, whenever possible. But then another question arises: if user buys products A and B, and there is one shop that could ship both, but 100 miles away from the customer, and two shops each having only one of the items, but within 10 miles of the customer, what solution should we choose? What about choosing between a closer shop that has a longer delivery time or higher cost, and another one further away but cheaper and ultimately better for the customer?

Even more issues come up if we lay down the architecture of the system (see figure 11.3): so far, we have treated shops datasets as if they were local, but that's not necessarily the